

Topic 7.3

7.3: Private Derivation

In a *public* derivation, public members in the base class are accessible from objects of the derived class. There are times when this provides too much access.

In a *private* derivation, public members in the base class are accessible only from the member functions of the derived class, not from any objects of the derived class.

The following example illustrates all possible cases. The access with the *blue italics* comment is the access that private derivation prevents.

```

Class Base
{
    private:
        int  base_private;

    protected:
        int  base_protected;

    public:
        int  base_public;
}

class Publicly_Derived : public Base
{
    private:
        int  derived_private;

    protected:
        int  derived_protected;

    public:
        int  derived_public;

    void member_function();
}

void Publicly_Derived::member_function()
{
    derived_private = 0; // All OK: we're
    // a member function.
    derived_protected = 0;
    derived_public = 0;

    //base_private = 0;    // Forbidden
    base_protected = 0;
    base_public = 0;
}

class Privately_Derived : private Base
{
    private:
        int  derived_private;

    protected:
        int  derived_protected;

```

```

    public:
        int derived_public;

    void member_function();
}

void Privately_Derived::member_function()
{
    derived_private = 0; // All OK: we're
    // a member function.
    derived_protected = 0;
    derived_public = 0;

    //base_private = 0;    // Forbidden
    base_protected = 0;
    base_public = 0;
}

int main(void)
{
    Base          b;
    PubliclyDerived pubd;
    PrivatelyDerived privd;

    //b.base_private = 0;    // Forbidden.
    //b.base_protected = 0;  // Forbidden.
    b.base_public = 0;      // OK.

    pubd.derived_public = 0;    // OK.
    //pubd.derived_protected = 0; // Forbidden.
    //pubd.derived_private = 0;  // Forbidden.
    pubd.base_public = 0;      // OK: public derivation.
    //pubd.base_protected = 0;   // Forbidden.
    //pubd.base_private = 0;    // Forbidden.

    privd.derived_public = 0;    // OK.
    //privd.derived_protected = 0; // Forbidden.
    //privd.derived_private = 0;  // Forbidden.
    //privd.base_public = 0;      // Forbidden:
    // private derivation.
    //privd.base_protected = 0;   // Forbidden.
    //privd.base_private = 0;    // Forbidden.

    return 0;
}

```

OK. Those are the rules for private derivation. So what? Well, to lead up to a realistic example, consider the blitz tour through graph theory coming up next (it's OK if you don't understand all or even most of this).

Topic 7.4 (<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-7-dot-4>)